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More Data Processing with Pandas

Week 3 - Assignment

Quiz: Quiz 3 30 min

Programming Assignment: Assignment 3 3h

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1. Consider the two DataFrames shown below. Which of the following expressions can be used to get the data of all students (from student_df) including their roles as staff, where nan denotes no role? 1 / 1 point

student_df

	School
Name	
James	Business
Mike	Law
Sally	Engineering

staff_df

	Role
Name	
Kelly	Director of HR
Sally	Course liasion
James	Grader

Your grade

100%

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- ☐ pd.merge(student_df, staff_df, how='right', left_index=True, right_index=True)
- ☐ pd.merge(staff_df, student_df, how='left', left_index=True, right_index=True)
- ☒ pd.merge(student_df, staff_df, how='left', left_index=True, right_index=True)
- ☐ pd.merge(staff_df, student_df, how='right', left_index=False, right_index=True)
- ☒ Correct
Using pd.merge() will select the first DataFrame as the left table and the second DataFrame as the right table. In order to get all records in the student_df, we can put it on the left side of 'left' join.

2. Consider a DataFrame named df with columns named P2010, P2011, P2012, P2013, 2014 and P2015 containing float values. We want to use the apply method to get a new DataFrame named result_df with a new column AVG. The AVG column should average the float values across P2010 to P2015. The apply method should also remove the 6 original columns (P2010 to P2015). For that, what should be the value of x and y in the given code? 1 / 1 point

```
3 result_df = df.drop(frames,axis=y)
```

- ☒ x = 1
y = 1
- ☐ x = 1
y = 0
- ☐ x = 0
y = 1
- ☐ x = 0
y = 0
- ☒ Correct
axis = 1 represents columns and axis=0 (the default) represents rows. Since frames represents all column titles, both methods need to act on columns, so both x and y will be 1

3. Consider the DataFrame df below, instantiated with a list of grades, ordered from best grade to worst. Which of the following options can be used to substitute X in the code given below, if we want to get all the grades between 'A' and 'B' where 'A' is better than 'B'? 1 / 1 point

```
1 import pandas as pd
2 df = pd.DataFrame(['A+', 'A', 'A-', 'B+', 'B', 'B-', 'C+', 'C', 'C-', 'D+', 'D'], index=['excellent', 'excellent', 'excellent', 'good', 'good', 'good', 'ok', 'ok', 'ok', 'poor', 'poor'], columns = ['grades'])
3 my_categories= X
4 grades = df['grades'].astype(my_categories)
5 result = grades[(grades>'B') & (grades<'A')]
```

- ☐ (my_categories=['A+', 'A', 'A-', 'B+', 'B', 'B-', 'C+', 'C', 'C-', 'D+', 'D'], ordered=True)
- ☒ my_categories = pd.CategoricalDtype(categories=['D', 'D+', 'C-', 'C', 'C+', 'B-', 'B', 'B+', 'A-', 'A', 'A+'], ordered=True)
- ☐ my_categories = pd.CategoricalDtype(categories=['D', 'D+', 'C-', 'C', 'C+', 'B-', 'B', 'B+', 'A-', 'A', 'A+'])
- ☐ my_categories = pd.CategoricalDtype(categories=['A+', 'A', 'A-', 'B+', 'B', 'B-', 'C+', 'C', 'C-', 'D+', 'D'])
- ☒ Correct
For the inequality in result to work, the list my_categories needs to be ordered which can be done using CategoricalDtype.

4. Consider the DataFrame df shown in the image below. Which of the following can return the head of the pivot table as shown in the image below df? 1 / 1 point

df

	world_rank	institution	country	Rank_Level
0	1	Harvard University	USA	First Tier Top University
1	2	Massachusetts Institute of Technology	USA	First Tier Top University
2	3	Stanford University	USA	First Tier Top University
3	4	University of Cambridge	United Kingdom	First Tier Top University
4	5	California Institute of Technology	USA	First Tier Top University

pivot table

	median				
Rank_Level	First Tier Top University	Other Top University	Second Tier Top University	Third Tier Top University	All
country					
Argentina	NaN	44.390	NaN	NaN	44.390
Australia	48.056	44.560	49.126	47.296	44.765
Austria	NaN	44.630	NaN	47.030	44.690
Belgium	51.375	44.715	49.630	46.890	46.210
Brazil	NaN	44.365	49.565	NaN	44.390

- ☐ df.pivot_table(values='score', index='country', columns='Rank_Level', aggfunc=[np.median])
- ☐ df.pivot_table(values='score', index='Rank_Level', columns='country', aggfunc=[np.median])
- ☐ df.pivot_table(values='score', index='Rank_Level', columns='country', aggfunc=[np.median], margins=True)
- ☒ df.pivot_table(values='score', index='country', columns='Rank_Level', aggfunc=[np.median], margins=True)
- ☒ Correct
In the pivot table, the column 'country' is the index (not Rank_Level) and 'margins=True' must be there to get the 'All' column added to the pivot_table.

5. Assume that the date '11/29/2019' in MM/DD/YYYY format is the 4th day of the week, what will be the result of the following? 1 / 1 point

```
1 import pandas as pd
2 (pd.Timestamp('11/29/2019') + pd.offsets.MonthEnd()).weekday()
```

- ☐ 7
- ☒ 5
- ☐ 6
- ☐ 4
- ☒ Correct
The result would be the end date of the month, which is exactly the next day of the given date - 11/30/2019. So when we call the weekday() on the resultant pd.Timestamp, it will be 5.

6. Consider a DataFrame df. We want to create groups based on the column group_key in the DataFrame and fill the nan values with group means using: 1 / 1 point

```
1 filling_mean = lambda g: g.fillna(g.mean())
```

Which of the following is correct for performing this task?

- ☐ df.groupby(group_key).transform(filling_mean)
- ☒ df.groupby(group_key).apply(filling_mean)
- ☐ df.groupby(group_key).filling_mean()
- ☐ df.groupby(group_key).aggregate(filling_mean)
- ☒ Correct
This is correct as the apply() function can be used to apply a function along an axis of a DataFrame.

